Paper Review

Title:

Preliminary Investigation of Spatial Variability of Sediment Fingerprinting Properties in

Relation to Terrain Attributes at the Field-Scale

The paper utilizes data collected from two field sites in the Manitoba Province, Canada, to investigate how easily measurable and remotely sensed terrain properties can predict soil fingerprinting properties. The work uses widely used methods for data collection, which are then analysed using rigorous statistical techniques. It shows that simple terrain properties as elevation and Wetness Index are among the best explanatory variables for most geochemical and colour fingerprinting properties, and that, therefore, for the forested site, a more clearly defined drainage network leads to clearer clustering of soil properties in terms of principal components.

Overall, the work is rigorous, uses accepted methods, and presents clear results. Nonetheless, the discussion is quite limited and could be improved by providing general rules (or the extent to which the presented conclusions could be generalized) for the selection of sampling locations in soil fingerprinting studies.

My suggestion is to accept with minor revisions.

Detailed comments and suggestions for the authors are provided below.

Title

I would suggest removing the preliminary, since the investigation should either be complete before publishing, or wait for its completion to derive conclusions. I would also suggest to add the area of the study. A suggestion might be “Investigation on the spatial variability of sediment fingerprinting properties in relation to terrain attributes in two field sites, Manitoba Province, Canada”

Abstract

The abstract is relatively clear but cumbersome in reading, as it does not balance well between important and almost superfluous details. I would suggest the authors to rewrite it to clearly state the main objective of their work. There is no reason to mention all 15 properties in the abstract, but never mention the case studies, because the results of this work are very specific to the study location and their generalization is limited. This is also a factor that could be made more explicit both in the abstract and in the conclusions. Overall, I would suggest focusing more on the lessons learnt from this study in terms of the relative importance of terrain properties on soil properties and trying to explain the physical/chemical reasons behind it. With respect to the objectives, they seem to be the result of modifications from an original version to accommodate some reviewers' requests. My personal suggestions are:

1) In objective one, the term spatial *dependence* is used, but it does not mention what those properties are supposed to depend upon spatially. Possibly, an introductory statement would make things easier. Something like: “This work uses data collected at two sites in Manitoba Province, Canada to assess …… . A set of 15 properties was used as fingerprints:….. Their spatial variability was studied and their dependency on a set of seven terrain attributes:…. was assessed through correlation and RF.”
2) In my previous suggestion, I merged objectives one and two, but obviously, the authors can choose to keep them separate if they so prefer. In this case, the part on spatial dependency should only be mentioned in objective 2.
3) The identification of patterns in the data is introduced but not justified. I would suggest adding just one sentence to explain why this might be important/useful.

L33: “in” is missing in “identify patterns **in** the data”.

Key findings

The first two key findings are very similar and can be joined. In the third key point, it is not clear what a soil predictor is.

1 Introduction

L49: I suggest considering changing “Developing a sampling design” to " Designing a sampling strategy” or protocol.

L71: A reference supporting this statement would be appreciated.

L72: The concept of representative sampling is presented without explaining what the authors mean by it. When can a sampling pattern be defined as representative? To what degree is representativeness a property of the sampling strategy only, and not a combined result of the inherent variability of the landscape and the fingerprinting methods used?

L75-80: I suggest including more references that support this argument. Examples could be:

Du, P., & Walling, D. E. (2017). Fingerprinting surficial sediment sources: Exploring some potential problems associated with the spatial variability of source material properties. *Journal of Environmental Management*, *194*, 4–15. https://doi.org/10.1016/j.jenvman.2016.05.066

Collins, A. L., Blackwell, M., Boeckx, P., Chivers, C.-A., Emelko, M., Evrard, O., et al. (2020). Sediment source fingerprinting: benchmarking recent outputs, remaining challenges and emerging themes. *Journal of Soils and Sediments*, *20*(12), 4160–4193. https://doi.org/10.1007/s11368-020-02755-4

Derakhshan-Babaei, F., Nosrati, K., Fiener, P., Egli, M., & Collins, A. L. (2024). Source fingerprinting sediment loss from sub-catchments and topographic zones using geochemical tracers and weathering indices. *Journal of Hydrology*, *633*, 131019. https://doi.org/10.1016/j.jhydrol.2024.131019

L82-84: It is not clear when an input is well defined, especially in relation to the next sentence stating that their spatial variability is not considered.

L84-86: The argument according to which samples closer to a stream might better represent a source appears unclear. My interpretation of the authors' sentence is that the contribution of areas closer to a stream might be higher and therefore represent a disproportionate fraction of the transport. If this is the case, it should be made more explicit; otherwise, the concept should be more clearly explained. Moreover, a reference supporting this argument could also be beneficial.

L104: Why would the analysis of a large number of elements be linked to the challenges due to limited data on soil properties? This sentence is not fully clear to me.

L111-114: Several concepts need to be linked more explicitly. Why is the wide adoption of DEM-based analysis to define sampling locations linked to the need for characterizing relationships with terrain attributes? Logic would suggest that one should look for relationships between landscape properties (DEM-based) and the spatial distribution of previously mentioned factors affecting soil properties: weathering, erosion, geology, etc… All the above was used just to introduce Random Forest, which is only one of the possible regression techniques. Overall, in the introduction, it is not explained why these regression techniques were chosen.

L142: It is not explained why such clustering methods were chosen.

2 Methods

L156: I suggest using % as an alternative unit for the creek slope, as m m-1 does not read well. This is just a personal suggestion to the authors.

Fig. 1 b) is not of good resolution, while c) is superfluous. I would suggest removing c) and enlarging and improving the quality of b). The relief of the two sites can be appreciated in d) and e). Sampling sites (and possibly their spacing) are not included in any legend.

2.2 Soil sampling and analysis

Data were not made public to allow reproducibility of the results. I would strongly suggest including them in an open data sharing platform (e.g. PANGAEA, Zenodo, Figshare or similar).

L168: Litter, Fermentation, Humus Layer (LFH) acronym is presented but not explained.

L169: I suspect that agricultural samples were "collected" 0-15 cm "below or deep".

L186-187: "detailed" is repeated.

L198: r^2 is mentioned but not reported, even as Supporting Information.

L205-208: It is not clear why, if both native DEMs were resampled at 1 m resolution, these were then resampled again at 10 m resolution? Moreover, the description of this paragraph first explains which attributes were computed and then returns to the topic of spatial resolution, which was already discussed when mentioning the 1 m resampling. I would suggest explaining which were the native resolutions, which resampling was done and why, and finally, which attributes were derived.

L213: I imagine that CV stands for coefficient of variation, but this is not made explicit.

L228: “using” is repeated.

L229: A parenthesis should be removed after “stable” or added before it.

At lines 225-226, it is said that the model performance was assessed using MSE and percentage variance explained, while at lines 229-231, R^2 is also mentioned. Please make it coherent.

3 Results

L 215: “are” is missing after properties.

L267: Figure 2 colorbars have no units, it is not clear what is shown there. Previously (L264) the nugget of the semivariogram is mentioned, but it is not clearly expressed in the following text.

L293: Figure 4- The indication of (% Increase in Mean Squared Error) below the graph might be misleading. I imagine that what matters is the Importance ranking, which coherently means that 1 is the most important, not that it represents the variable with 1% in RMSE increase. I would suggest removing the text in parentheses.

4 Discussion

L314-340: This section resembles more an introduction section than a discussion, as it lays the context for the discussion. I suggest the authors to consider merging it in terms of content (not copy/pasting) with the introduction, to shorten the overall manuscript, improve legibility, and allow the reader to focus more on discussions. This is also supported by the fact that it is not smoothly linked to the following paragraph.

Overall, the discussion in section 4.1 is more a summary of the quantitative findings, than a discussion linking processes to observed data. Moreover, I miss the reason why there should be the need for two separate sections: 4.1 Variability of soil properties, and 4.22 Spatial distribution, as there is no other variability, other than spatial, discussed in the paper (time factors are not considered). I admit that I might be missing a point here; I would appreciate an explanation from the authors.